IIT Kharagpur

Subhajit Mondal <subhajit2005@kgpian.iitkgp.ac.in>

LLM-driven intelligent procurement assistant

3 messages

mj@iem.iitkgp.ac.in <mj@iem.iitkgp.ac.in> Wed, Jan 14, 2026 at 2:44 PM
To: mudiit.raj@kgpian.iitkgp.ac.in, subhajit2005@kgpian.iitkgp.ac.in
Cc: Sahil Kaushik <sahil.kaushik123987@gmail.com>, Falguni Mahato <falguni@iem.iitkgp.ac.in>, amit1539singh@gmail.com

LLM-Driven Intelligent Procurement Assistant

23IM10036	Subhajit Mondal	sm829937@gmail.com
23IM30015	Mudit Raj	dktanti66@gmail.com

Problem Statement
LLM-Driven Intelligent Procurement Assistant for an Automotive Components Manufacturer
An automotive components manufacturing company (Tier-1/Tier-2 supplier) procures a wide range of direct and indirect materials, including machined parts, castings, fasteners, electronics, packaging, maintenance spares, and contract services. Procurement activities involve drafting RFQs, interpreting technical specifications and drawings, shortlisting qualified vendors, comparing bids, checking compliance with quality requirements (e.g., IATF 16949), evaluating delivery and risk terms, and preparing purchase recommendations for internal approvals. Much of this work is document-heavy and time-consuming, and critical knowledge is scattered across RFQs, emails, vendor profiles, historical purchase orders, contracts, and quality/audit reports. Currently, the procurement team relies on manual review and spreadsheet-based comparisons. This results in long RFQ cycles, inconsistent vendor shortlisting, missed clause risks in quotations/terms, limited reuse of historical knowledge, and weak traceability of decision rationale during audits. Procurement managers need a decision-support information system that can extract structured information from unstructured procurement documents, compare bids consistently, highlight compliance gaps and contractual risks, and generate clear procurement recommendations in managerial language—while remaining auditable and under human control.
The objective of this project is to design and implement an **LLM-driven Intelligent Procurement Assistant Information System** for an automotive components manufacturer. The system will ingest RFQs, vendor quotations, and related documents; extract key procurement attributes into structured templates; support vendor shortlisting and bid comparison; flag compliance and contractual risks using rule checks; and use an LLM to generate evidence-backed procurement summaries and follow-up questions. The system must include a reviewer workflow (approve/reject/on-hold), maintain a complete audit trail, and avoid hallucinations by grounding all generated content in extracted evidence and computed comparisons.

Expected Deliverables

1. **Procurement Workflow and System Architecture**
   End-to-end workflow (RFQ → vendor response → evaluation → recommendation → approval) and architecture showing ingestion, extraction, analysis, LLM layer, UI, and audit log.
2. **Data Model and Templates**
   Structured schema for RFQ metadata, item specifications, vendor profiles, bid line items, terms, compliance checks, scoring, and decision records.
3. **Document Ingestion and Extraction Module**
   Upload and parse RFQs/quotations/contracts; extract fields such as part number, quantity, delivery lead time, unit price, incoterms, payment terms, warranty, certifications, and deviations.

4. **Vendor Shortlisting and Similarity Search Module**
   Match vendors to RFQ requirements using vendor capability profiles, past performance indicators, and similarity search over descriptions (RAG-style retrieval optional).
5. **Bid Comparison and Scoring Engine**
   Create a normalized bid comparison table; compute scores based on cost, lead time, quality compliance, risk flags, and historical performance (rule-based weighting acceptable).
6. **Compliance and Contract Risk Flagging Module**
   Company-specific checks (e.g., IATF certification, PPAP readiness where relevant, delivery penalties, warranty terms, liability clauses, non-standard payment terms).
7. **LLM Recommendation and Clarification Generator**
   Generate an evidence-based procurement recommendation summary, highlight trade-offs, and draft clarification questions to vendors—strictly grounded in extracted data and scores.
8. **Reviewer Interface (Decision Support UI)**
   UI to upload documents, view extracted RFQ/bid templates, comparison tables, risk flags, LLM summary, and take decisions with comments.
9. **Audit Trail and Governance Layer**
   Log document versions, extraction results, scoring outputs, LLM prompt/output (or summaries), reviewer decisions, and overrides.
10. **Synthetic Procurement Dataset and Test Cases**
    Synthetic RFQs and vendor quotations with controlled variations (missing certifications, clause deviations, price-lead time trade-offs) for systematic testing.
11. **Final Report and Demonstration**
    Architecture, workflow, sample documents/templates, evaluation results, limitations, and demo.

Weekly Work Plan (Two Students)

Week	Student A – Extraction, Scoring & Backend Logic	Student B – LLM Layer, UI & Governance	Joint / Integration Focus
1	Study automotive procurement workflow; choose 2–3 RFQ categories (e.g., machined part, packaging, service)	Identify user roles (buyer, procurement manager, quality); define system outputs	Finalize company context and RFQ scope
2	Design data model: RFQ, vendor, bid items, terms, checks, decisions	Draft architecture: ingestion → extraction → scoring → LLM → UI → logs	Freeze schema + architecture diagram
3	Implement document ingestion and storage; build extraction templates	Build UI skeleton (RFQ upload, project list, vendor list)	Connect UI upload to backend storage
4	Implement extraction for key bid fields (price, lead time, MOQ, delivery, terms)	Define LLM input/output schema; prompt templates	Align extracted fields with LLM grounding
5	Build bid comparison table generator; normalization rules	Implement LLM summary: bid comparison narrative + trade-offs	First end-to-end run on sample RFQ
6	Implement scoring engine (cost, lead time, compliance, risk)	Implement reviewer actions (shortlist/hold/reject + comments)	Integrate scoring + decision workflow
7	Implement compliance/risk flag rules (certs, payment, warranty deviations)	Add guardrails: cite evidence; no numeric hallucination; structured output	Stress-test on tricky clause variations
8	Build vendor matching module (capability tags + similarity search)	Integrate vendor shortlist view into UI	Joint testing: shortlist + comparison
9	Create synthetic RFQs and quotations dataset; edge cases	Add audit trail screens and exportable evaluation report	Run systematic test suite
10	Improve extraction robustness; handle multi-page docs and missing fields	Refine LLM outputs for clarity and managerial usefulness	Integration freeze and bug fixing
11	Evaluate: extraction accuracy, scoring stability, cycle time reduction (simulated)	Prepare demo script and screenshots; usability feedback	Draft final report sections
12	Finalize report: data model, scoring/rules, evaluation	Finalize report: LLM design, UI, governance/audit	Final demo and submission

--
With regards
Mamata Jenamani
Professor
Department of Industrial and Systems Engineering
IIT Kharagpur

Subhajit Mondal <subhajit2005@kgpian.iitkgp.ac.in> Wed, Jan 14, 2026 at 11:49 PM
To: "rishi125.works.here@gmail.com" <rishi125.works.here@gmail.com>

[Quoted text hidden]

Subhajit Mondal <subhajit2005@kgpian.iitkgp.ac.in> Thu, Jan 15, 2026 at 3:34 AM
To: "ssudarshanpandey@gmail.com" <ssudarshanpandey@gmail.com>

---------- Forwarded message ---------
From: <mj@iem.iitkgp.ac.in>
Date: Wed, 14 Jan, 2026, 2:45 pm
Subject: LLM-driven intelligent procurement assistant
To: <mudiit.raj@kgpian.iitkgp.ac.in>, <subhajit2005@kgpian.iitkgp.ac.in>
Cc: Sahil Kaushik <sahil.kaushik123987@gmail.com>, Falguni Mahato <falguni@iem.iitkgp.ac.in>, <amit1539singh@gmail.com>

[Quoted text hidden]